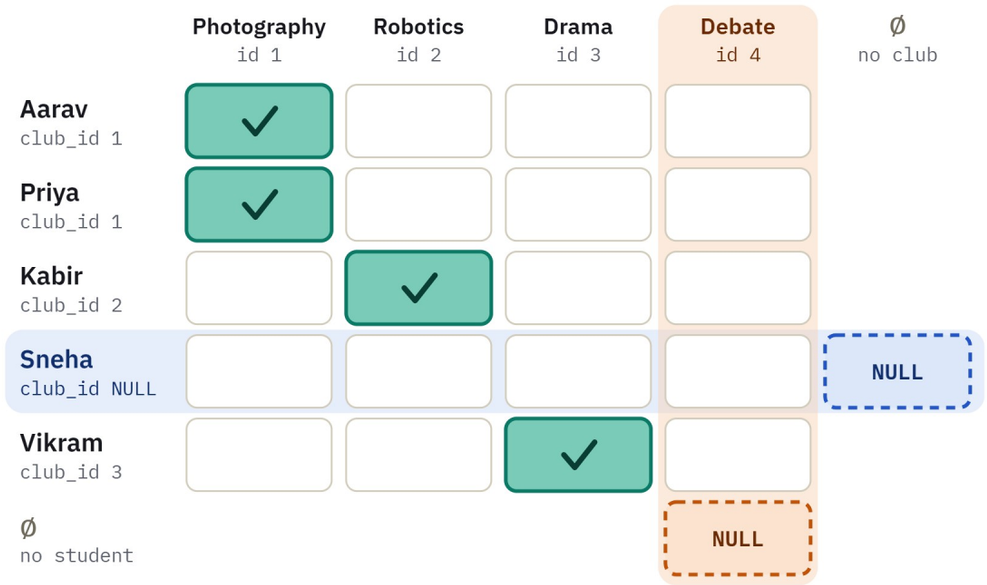
Every join is a filtered grid.

Rows are students, columns are clubs, each cell is one possible pair. A join decides which cells to keep.

5 STUDENTS × 4 CLUBS



1 **CROSS JOIN = the whole grid**

No ON. Every student paired with every club: 5 × 4 = **20 rows**.

2 **ON club_id = club_id = light up cells**

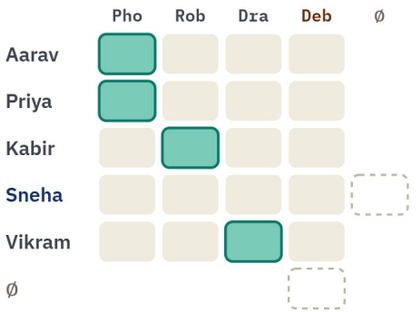
Only the cells where the ids agree stay lit. Those **4 cells are INNER JOIN**.

3 **Outer joins rescue empty lines**

Sneha's row and **Debate's column** have no lit cell. The dashed `NULL` cell pairs an orphan with nothing so it survives.

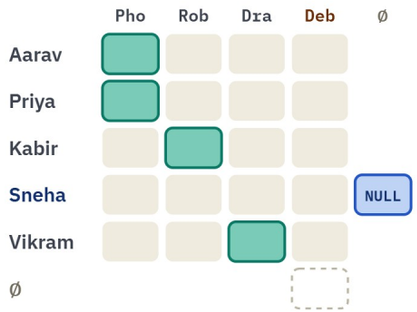
ids match · pair exists, no match · NULL rescue

INNER — 4 ROWS



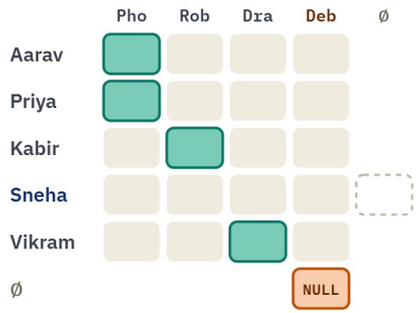
Lit cells only. Both orphans are gone.

LEFT — 5 ROWS



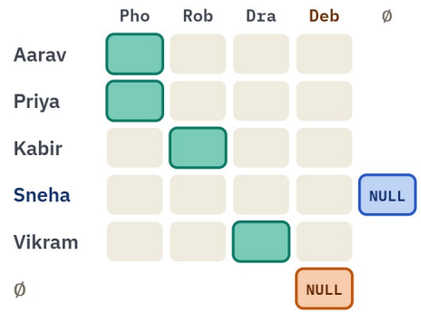
Lit cells + **Sneha**'s NULL cell.

RIGHT — 5 ROWS



Lit cells + **Debate**'s NULL cell.

FULL — 6 ROWS



Lit cells + both NULL cells.